Truth Tables

October 9, 2024

Truth Table 1

Truth table for:

$$(\sim X \wedge \sim(\sim X \wedge \sim Y))$$

X	Y	$(\sim X \wedge \sim(\sim X \wedge \sim Y))$
1	1	0 0 1 0 0 0
1	0	0 0 1 0 0 1
0	1	1 1 1 1 0 0
0	0	1 0 0 1 1 1
		2 1 3 5 4 6

Truth Table 2

Truth table for:

$$\sim(X \vee \sim(X \vee Y))$$

X	Y	$\sim(X \vee \sim(X \vee Y))$
1	1	0 1 0 1
1	0	0 1 0 1
0	1	1 0 0 1
0	0	0 1 1 0
		1 2 3 4

Truth Table 3

Truth table for:

$$\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim(X \supset (Y \equiv X))$$

X	Y	$\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim\sim(X \supset (Y \equiv X))$
1	1	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
1	0	1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0
0	1	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
0	0	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
		1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Truth Table 4

Truth table for:

$$(\sim X \vee Y)$$

X	Y	$(\sim X \vee Y)$
1	1	0 1
1	0	0 0
0	1	1 1
0	0	1 1
		2 1

Truth Table 5

Truth table for:

$$(X \equiv (Y \supset Z))$$

X	Y	Z	$(X \equiv (Y \supset Z))$
1	1	1	1 1
1	1	0	0 0
1	0	1	1 1
1	0	0	1 1
0	1	1	0 1
0	1	0	1 0
0	0	1	0 1
0	0	0	0 1
			1 2

Truth Table 6

Truth table for:

$$(((X \supset Y) \wedge (X \supset \sim Y)) \equiv \sim X)$$

X	Y	$(((X \supset Y) \wedge (X \supset \sim Y)) \equiv \sim X)$
1	1	1 0 0 0 1 0
1	0	0 0 1 1 1 0
0	1	1 1 1 0 1 1
0	0	1 1 1 1 1 1
		3 2 4 5 1 6

Truth Table 7

Truth table for:

$$((X \wedge Y) \equiv (X \equiv (X \supset Y)))$$

X	Y	$((X \wedge Y) \equiv (X \equiv (X \supset Y)))$			
1	1	1	1	1	1
1	0	0	1	0	0
0	1	0	1	0	1
0	0	0	1	0	1
		2	1	3	4